

TahamajaNet: A 72B→7B Distillation Pipeline with Cardinality-Aware Fine-Tuning and an End-to-End Semantic Metric for Cafeteria Tray Analysis

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Abstract

We present **TahamajaNet**, a three-stage detection pipeline that turns a single cafeteria-tray image into the multiset of food items needed to generate a bill, and we identify the central failure mode of generative-VLM detection on this task: *for VLM-as-detector, syntax and localization are not the bottleneck; cardinality is*. We make four contributions. (i) **A 72B→7B distillation pipeline**: a 72B teacher labels 4,999 trays once offline (Qwen2.5-VL-72B-AWQ → SAM3 → DINOv2/SigLIP cluster review) and a 7B student runs the same three stages online. (ii) **A cardinality-aware structural loss** that addresses an asymmetric overcount regression we measure quantitatively: matched bbox IoU climbs from 0.32 to 0.90 across training while exact-count accuracy drops from 86.8% to 53.5%. (iii) **Dish-Correct**, an end-to-end semantic metric (count-exact $\wedge$ class-multiset-match) that aligns with bill correctness and forgives bbox geometry. (iv) **The v3.2 dataset**, a 4,629-item, 32-class, hand-curated Saudi cafeteria-tray dataset, released with splits. On 596 held-out images, the distilled 7B reaches **86.4%** dish-correct, a **+47.0 pp** absolute lift over the off-the-shelf 7B baseline (zero-detection rate 37.8% → 0.0%, per-item recall 45.4% → 90.6%).

1 Introduction

Generating an itemized bill from a single photo of a cafeteria tray is a deceptively hard vision problem. The output is not a set of pixel-accurate masks, nor a ranked list of detection scores: it is a *multiset of food classes* priced by a downstream lookup. A single missed item or a single duplicated detection is a wrong bill. A bbox that overlaps the ground truth at IoU 0.5 versus 0.9 is, at the level of revenue, indistinguishable.

This semantic objective interacts poorly with two off-the-shelf options. Large vision-language models (VLMs) at the 70B parameter scale will reliably *describe* a tray image but are too slow and too expensive for inference on every dish (target latency: < 2000 ms per image). Small open-set detectors are fast but lack the dish-level concept ontology required to merge top-

pings with their base dish, separate co-located items, and ignore plates and utensils.

We bridge these regimes with four contributions.

(1) **A 72B→7B distillation pipeline (§3, §4)**. A Qwen2.5-VL-72B-AWQ teacher labels 4,999 cafeteria images via a structured-JSON prompt, followed by SAM3 segmentation, DINOv2 + SigLIP clustering, and human cluster curation. The resulting 4,629-item dataset (32 classes) trains a Qwen2.5-VL-7B student via LoRA on the language model and an unfrozen vision encoder. The teacher pipeline runs once, offline; the student runs at every inference.

(2) **Cardinality-aware fine-tuning (§5)**. We observe a clear *cardinality-localization tradeoff*: standard supervised fine-tuning improves bbox geometry while degrading exact instance count, with overcount the dominant error mode. We introduce a structural count loss that penalises $|\hat{n} - n|$ during training and arrests the regression. Our central claim: *for generative VLM-based detection, syntax and localization are not the bottleneck; cardinality is*.

(3) **Dish-Correct (§6)**. We define an end-to-end semantic evaluation metric whose hierarchy — count-exact, class set-IoU, class multiset match, dish-correct — aligns with the bill-generation objective and forgives bbox geometry. We argue per-stage IoU is a misleading proxy in this regime, and we report all results on Dish-Correct.

(4) **The v3.2 dataset, public release**. A 4,629-item, 32-class hand-curated Saudi cafeteria-tray dataset — the first of its scale and domain that we are aware of. The release reflects significant compute (96-minute Qwen-VL-72B vLLM run on 80 GB GPUs, plus DINOv2-large + SigLIP embedding for ~5,000 crops, plus HDBSCAN over the joint embedding) and significant human time: a curator inspected and labeled *every cluster centroid*, collapsing 467 raw HDBSCAN clusters into the 32 production classes, and recovered 1,344 noise-bin items via visual review. Train / dev / test splits and the canonical 80/10/10 seed are part of the release.

2 Related Work

VLM-based detection. Recent instruction-tuned vision-language models (Qwen-VL, Qwen2.5-VL, LLaVA, InternVL, Florence-2, PaLI-X) cast bounding-box generation as a JSON-output token-prediction task: the model emits a structured response listing detected regions with coordinates and optional descriptors. This paradigm is attractive because the same prompt template generalises across object categories without needing a fixed-class detection head, and because the resulting boxes can be coupled with language-grounded captions in a single forward pass. The trade-off is twofold: (i) there is no explicit objectness prior over the predicted *number* of items, and (ii) bbox coordinates are emitted as text tokens that compete for attention with the region-naming tokens. We inherit both of these trade-offs and address (i) directly with a structural count loss (§5); (ii) we sidestep by training under a *bbbox-only* schema that strips region names from the supervision target, leaving fine-grained classification to a downstream embedding head.

Knowledge distillation. The bulk of distillation literature — Hinton’s response-matching, FitNets feature-matching, attention transfer, self-distillation — assumes a single *model* teacher whose internal states or output distributions are aligned with those of a student. Our setup is structurally different: the 72B teacher is a *labeling pipeline*, and the supervision target is the teacher’s deterministic post-processed output (class-collapsed bboxes after SAM3 mask refinement and human cluster review) rather than soft logits. There is no hidden-state match, no temperature-softmax KL term, no auxiliary loss on intermediate features. This makes the teacher swap-replaceable: any labeling pipeline that produces structured-JSON detections at acceptable quality could substitute for Qwen-VL-72B-AWQ. The price we pay is that we cannot exploit the teacher’s uncertainty signal — a soft logit might have told us *this is 60% rice, 40% biryani*, and our cluster-collapsed hard label cannot. We treat the resulting label noise as in-distribution.

Multi-stage detection pipelines. Two-stage R-CNN-family detectors (Faster R-CNN, Mask R-CNN) decouple region proposal from classification. End-to-end DETR and its descendants (Deformable DETR, DINO) use a single transformer over learned object queries. SAM and SAM2/SAM3 separated mask refinement from prompt generation, enabling compositional pipelines like Grounded-SAM (open-vocabulary box-from-text $\rightarrow$ SAM mask). Our pipeline is closest in spirit to Grounded-SAM but inverts the roles: the

box generator is fine-tuned for a closed taxonomy, the segmenter is frozen, and the classification step is explicitly retrieval (cosine over a fixed reference bank) rather than a learned head. The key novelty is the *symmetric* reuse of the same three-stage decomposition at both train-time labeling (72B) and inference (7B), at $\sim 10\times$ different scales — which makes the train/test distribution mismatch easier to reason about and lets us re-use infrastructure (the SAM3 segmenter, the DINOv2 vector DB) between the two phases verbatim.

Open-vocabulary and food-specific detection. Food recognition benchmarks (Food-101, Food-2K, ETHZ Food-101) are predominantly classification, not detection. The few detection-grade food datasets (UEC-Food100, UNIMIB, ETH Mensa) are small (under 5,000 images each) and skewed toward European/Asian cuisines. None of them, to our knowledge, contain the class taxonomy or cuisine specificity required for Saudi cafeteria-tray analysis: kabsa rice, masakaah, asidah/Saudi porridge, chicken/meat koftas, hummus (cling-wrapped), and the wrapped/sealed dairy sub-categories absent from international datasets. Our v3.2 release fills this gap.

Cardinality-aware evaluation. Standard detection metrics — COCO mAP, mean IoU, F1@0.5 — reward bbox geometry and ranking quality but are blind to absolute item count. Recent work in dense crowd counting (MCNN, CSRNet) and instance-level industrial inspection has argued for cardinality-explicit objectives. Dish-Correct generalises this argument to the multi-class, multi-instance, item-level-billing regime: at the output of the pipeline, the only thing that matters is whether the predicted multiset of class labels matches the ground-truth multiset. Bbox geometry is just an internal coordinate, not a deliverable.

3 Data Engine: The Teacher Pipeline

The teacher labeling pipeline is itself a three-stage cascade, run *once* on a curated 4,999-image subset of cafeteria tray photos. The output is a labeled training set; no part of this pipeline runs at inference time. Figure 2 shows the full architecture — note the deliberate symmetry with the student pipeline (Figure 5): the same three-stage decomposition, with frozen 72B / DINOv2+SigLIP / HDBSCAN at the labeling scale and fine-tuned 7B / SAM3 / DINOv2 k -NN at the inference scale.

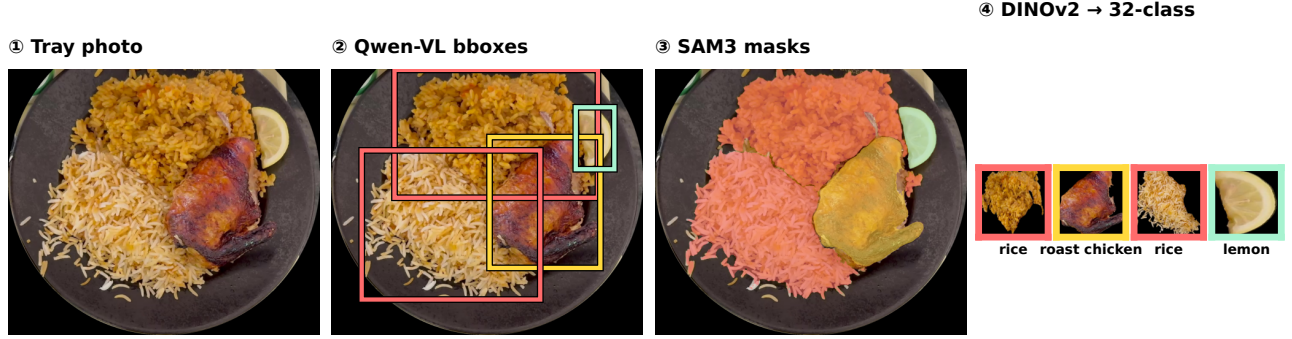


Figure 1: The teacher pipeline applied to a single tray. (1) Source image. (2) Qwen2.5-VL-72B emits a JSON list of bounding boxes. (3) SAM3 refines each bbox into a pixel mask. (4) Each masked crop is embedded with DINOv2-large + SigLIP, clustered with HDBSCAN, and collapsed by human curators into one of 32 dish classes. Here: rice, roast chicken, rice (second helping), lemon. The student inference pipeline (Figure 5) follows the same flow at smaller scale.

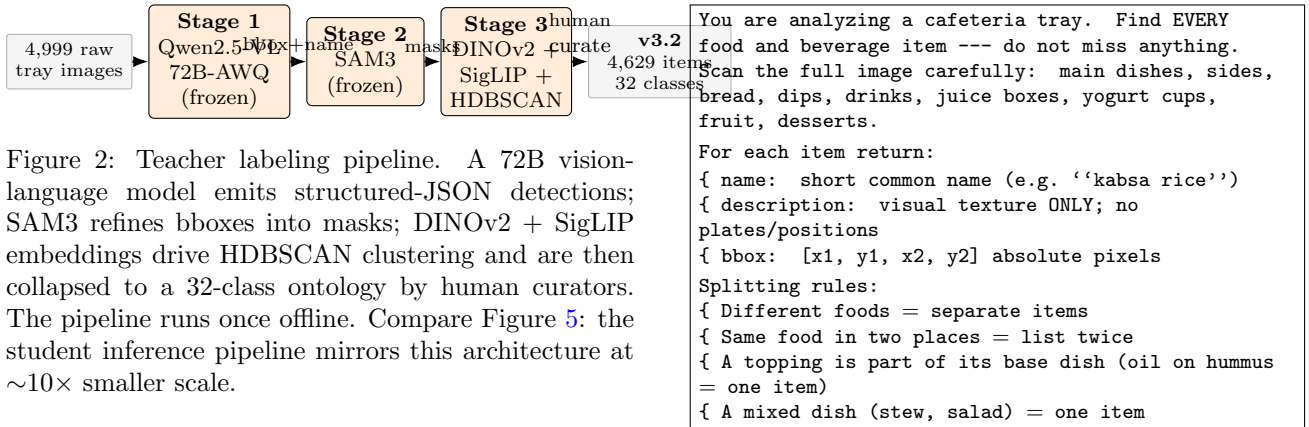


Figure 2: Teacher labeling pipeline. A 72B vision-language model emits structured-JSON detections; SAM3 refines bboxes into masks; DINOv2 + SigLIP embeddings drive HDBSCAN clustering and are then collapsed to a 32-class ontology by human curators. The pipeline runs once offline. Compare Figure 5: the student inference pipeline mirrors this architecture at $\sim 10\times$ smaller scale.

3.1 Stage 1 (Teacher VLM): Qwen2.5-VL-72B-AWQ

The first stage is a 72B-parameter vision-language model (Qwen/Qwen2.5-VL-72B-Instruct-AWQ) served via vLLM with FP8 KV-cache and prefix caching. It is prompted to emit a strict JSON list of **name**, **description**, and **bbox** per visible food region, with explicit *splitting rules* (mixed dishes are one item, toppings stay with their base dish) and *exclusion rules* (no plates, trays, utensils, empty containers, or labels). The full prompt — which encodes much of the dataset’s ontology — is reproduced in Figure 3.

The 72B teacher succeeds on 4,999 / 4,999 images (zero JSON-parse failures) in 5,778 s of wall-clock time (roughly 1.16 s/image at the AWQ-quantised weight precision).

Figure 3: Stage 1 prompt for the Qwen2.5-VL-72B teacher. The splitting rules encode the dataset’s ontology. *Toppings stay with their base* is the single most important constraint — it is what distinguishes “rice with sauce” (one item) from “rice and sauce” (two items) for billing purposes.

3.2 Stage 2 (Teacher SAM): SAM3 mask refinement

For each VLM-emitted bbox, SAM3 is prompted with the box and produces a pixel-accurate mask. We apply a confidence cascade ($0.10 \rightarrow 0.05 \rightarrow 0.02 \rightarrow 0.01$) so that low-contrast items are not silently dropped, and a global mask-IoU NMS at 0.7 to suppress duplicate masks within the same image. SAM3 succeeds on 4,948 / 4,999 images; the 51 failures (1.0%) are primarily very small or very low-contrast regions and are excluded from training rather than heuristically recovered.

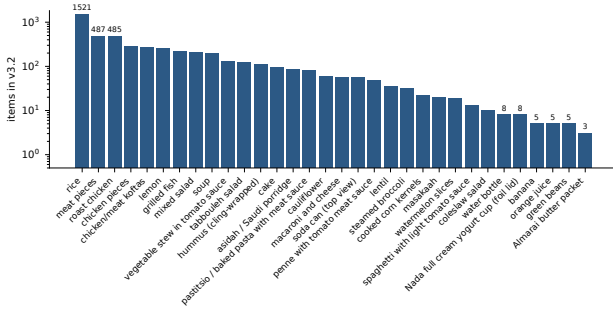


Figure 4: Per-class item counts in the v3.2 training set, sorted descending. The distribution is heavily long-tailed: top-3 classes account for over half of all items; five tail classes have fewer than ten items each.

3.3 Stage 3 (Teacher classifier): cluster + curate

The 72B teacher’s open-vocabulary names (e.g. “cheesecake slice”, “chocolate cake”, “yellow cake”) need to be collapsed into a closed taxonomy for downstream training. Each masked crop is embedded with frozen DINOv2-large (1024-d vision) and SigLIP (768-d, fused via the VLM’s `name` string), reduced via UMAP, and clustered with HDBSCAN. The result is 467 raw clusters, of which 1,744 items are labeled noise. Human curators inspect cluster centroids in an interactive viewer (`dataset_inspect/`) and (i) merge near-synonymous clusters (“cheesecake slice” + “chocolate cake” + “fruit tart” → **cake**), (ii) drop clusters that are not edible (e.g. utensil artifacts), and (iii) recover the highest-confidence noise items into the nearest valid cluster (the `noise_recovery` step in v3.1, surfacing 1,344 recovered items).

3.4 Final dataset (v3.2)

The released v3.2 export filters to images where every 72B-detected region survives SAM3 and clustering (i.e., complete labels per image). It contains **4,629 items across 2,981 unique images and 32 classes**, with an average of 1.55 items per image. Class frequency is heavily long-tailed (Figure 4): **rice** alone accounts for 32% of all items, while five tail classes have fewer than 10 items each. Items are partitioned into train/dev/test (80/10/10, image-level stratified, seed 420) before any model training.

3.5 Class taxonomy and distribution

The 32-class final ontology was reached by collapsing the 467 raw HDBSCAN clusters along three axes: (i) visual synonymy (different cheesecake variants → **cake**); (ii) price equivalence under the cafe-

Source images (deduplicated)	4,999
72B-VLM successful detections	4,999 (100.0%)
SAM3 successful segmentations	4,948 (99.0%)
Wall-clock teacher run-time	5,778 s ($\approx$ 96 min)
Raw HDBSCAN clusters	467
Noise items (HDBSCAN)	1,744
Final classes (post-curation)	32
Final items	4,629
Final unique images	2,981

Table 1: Teacher pipeline scale and yield.

Class	n	Class
rice	1521	cake
roast chicken	530	hummus (cling-wrapped)
meat pieces	515	pastitsio / baked pasta with meat sauce
chicken pieces	268	macaroni and cheese
lemon	240	lentil
mixed salad	210	penne with tomato meat sauce
soup	197	soda can (top view)
vegetable stew in tomato sauce	132	steamed broccoli
grilled fish	219	cauliflower
chicken/meat koftas	113	cooked corn kernels
tabbouleh salad	125	masakaah
asidah / Saudi porridge	86	watermelon slices
spaghetti with light tomato sauce	11	coleslaw salad
Nada full cream yogurt cup (foil lid)	7	Almarai butter packet
green beans	1	water bottle
orange juice	1	—

Table 2: All 32 v3.2 classes with item counts. The lower 6 rows hold the visible long-tail (single-digit support each); these classes are preserved in the dataset for completeness but our evaluation is conservative on them, and we flag any per-class result with $n < 10$ test instances as advisory only. The class names retain their original descriptors (e.g. “Almarai butter packet”) because billing requires the brand-specific SKU.

teria’s billing rules (**rice** subsumes plain rice, kabsa rice, mandi rice, biryani because they bill identically); and (iii) sub-cluster artefact removal (utensil-only or empty-plate clusters were dropped). Table ?? lists all 32 classes with their v3.2 item counts; the distribution is heavily long-tailed (top class contains 32% of all items, 5 tail classes have fewer than 10 items each).

3.6 Quality control and curator workflow

Cluster curation was performed in an interactive HTML viewer (`dataset_inspect/cluster_viewer.html`) that surfaces, for each of the 467 raw clusters, (i) the 16 closest crops to the cluster centroid in DINOv2+SigLIP joint space, (ii) the union of VLM-emitted class names within the cluster, and (iii) a free-text `display_name` input. A single curator labeled all 467 clusters in $\approx$ 4 hours, with a second pass to merge near-duplicate clusters (e.g. **rice** appearing as 8 small clusters be-

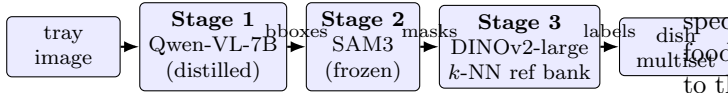


Figure 5: Student inference pipeline. Stage 1 is a fine-tuned 7B VLM that emits bboxes only (class-agnostic). Stage 2 (SAM3, frozen) refines each box into a pixel-accurate mask. Stage 3 (DINOv2-large) maps the masked crop into embedding space and assigns the class of the most similar of $\sim 3,900$ training-set reference embeddings. The teacher pipeline (§3) has the same three stages at $\sim 10\times$ scale, run once offline.

cause the 72B teacher emitted slight name variants). Boundary clusters — those with ambiguous content or visible mis-labeling — were resolved by inspecting 10 random crops outside the centroid neighbourhood; if the cluster was genuinely heterogeneous it was split, otherwise the dominant label was applied.

The HDBSCAN noise bin (originally 1,744 items, $\approx 35\%$ of the post-SAM yield) was processed by a separate *noise recovery* pass that re-embedded each noise item, computed cosine similarity to each curated cluster’s centroid, and assigned it to the nearest cluster if the similarity exceeded 0.75. This recovered 1,344 items (77% of noise) at the price of one parameter (0.75). Items that remained in the noise bin were dropped from v3.2 entirely. Per-cluster recovery threshold tuning is in scope for v3.3.

4 Student Pipeline: Tahama-jaNet Inference

The student inference pipeline mirrors the teacher’s three-stage decomposition at a smaller scale: the 72B VLM is replaced by a distilled 7B detector, the SAM3 segmenter is shared, and the offline DINOv2 + clustering step is replaced by an online DINOv2 vector retrieval head over the curated 32-class taxonomy. Figure 5 shows the full forward pass.

4.1 Stage 1: distilled Qwen2.5-VL-7B

The detector is `Qwen/Qwen2.5-VL-7B-Instruct` fine-tuned on v3.2 (§3) under a deliberately *class-agnostic* prompt schema:

```

Detect every visible edible food or beverage
region...
Format: {"items":[{"bbox": [x1,y1,x2,y2]}]}
Each item object must contain exactly one key:
bbox.

```

We deliberately strip class names and descriptions from the supervision target. This forces Stage 1 to

specialise in what VLMs do well — locating cohesive food regions — while leaving fine-grained classification to the embedding-based head in Stage 3, which scales more naturally with the long-tailed class distribution. Training uses LoRA ($r = 16$, $\alpha = 32$) on the language model and an unfrozen vision encoder; the latter recovers $\sim 691\text{M}$ parameters of the 7B that the LoRA-only path would otherwise leave frozen.

4.2 Stage 2: SAM3 (frozen, shared with teacher)

SAM3 is run with each Stage 1 bbox as a positive prompt. We retain the teacher pipeline’s confidence cascade and global mask- IoU NMS at 0.5. If SAM3 returns no kept masks (a $< 1\%$ failure mode), the pipeline falls back to bbox crops — the downstream classifier still produces a label but without background suppression.

4.3 Stage 3: DINOv2 maximum-similarity retrieval

Rather than a trainable classification head, Stage 3 is a *retrieval* problem over the train-split items. At setup, we embed every train-split masked crop with frozen DINOv2-large (1024-d, L2-normalised); call this the *reference bank* $\mathbf{R} \in \mathbb{R}^{N \times 1024}$ where $N \approx 3,900$. Each row $\mathbf{r}_i$ is annotated with its class c_i . At inference, the masked query crop q is embedded into the same space and we compute

$$\hat{c}(q) = c_{i^*} \quad \text{where} \quad i^* = \arg \max_i \langle \mathbf{R}_i, \mathbf{e}(q) \rangle.$$

Unlike the more common mean-pooled prototype (one vector per class), the k -NN top-1 formulation handles the multimodal classes that arise from human cluster curation — e.g. `rice` contains plain rice, kabsa rice, biryani, and mandi rice, each with a distinct visual signature. A mean-pooled prototype averages across these modes; max-similarity retrieval matches the query to the nearest concrete reference instance.

The reference bank is built from *train-split images only*. Held-out dev and test images never appear in $\mathbf{R}$, so the test-time cosine score cannot be inflated by a query matching its own reference.

4.4 Training setup

The Stage 1 fine-tune runs on a single H100 SXM (80 GB) at bf16 mixed precision under `HuggingFace transformers` ≥ 4.46 and `peft` ≥ 0.13 . Hyperparameters are summarised in Table ??.

		Stage	Latency (ms)
Base model	Qwen/Qwen2.5-VL-7B-Instruct		
LoRA rank, α	$r=16, \alpha=32$	Stage 1: Qwen-VL-7B detect (bbox-only JSON, fp16)	$\approx 1,800$
LoRA targets	q, k, v, o projections (all attn layers)	Stage 2: SAM3 mask refinement	≈ 500
LoRA dropout	0.05	Stage 3: DINOv2-large embed + max-sim retrieval	≈ 200
Vision encoder	unfrozen (~ 691 M params trained)		
Optimizer	AdamW, $\beta_1=0.9, \beta_2=0.999$	End-to-end (mean over 596 held-out trays)	$\approx 2,500$
Learning rate	2×10^{-5} peak, cosine to 1×10^{-6}		
Warmup	200 steps		
Batch size	4 (per-device), grad-accum 4 $\rightarrow$ 16 effective		
Sequence length	2048 (covers all v3.2 prompts)		
Image input	resized to mask-native (~ 1024 long-edge)		
Processor max_pixels	$1280 \times 28 \times 28 \approx 1,003,520$		
Schema	bbox-only (<code>{"items": [{"bbox": ...}]}</code>)		
Count loss weight λ	5.0 (cardinality-aware variant only)		
Epochs planned	10		
Wall-clock per epoch	≈ 25 minutes on H100 SXM		
Data	v3.2 train (2,385 images, 3,909 items)		
Validation	v3.2 dev (298 images), every epoch		
Checkpoint selection	best val EXACT-COUNT $\wedge$ MULTISSET-MATCH		
Determinism	cuDNN deterministic, seed 1337		

Table 3: Stage 1 fine-tuning hyperparameters. The LoRA targets, base model, and optimizer follow the standard recipe for instruction-tuned VLM fine-tuning; the non-default choices are the unfrozen vision encoder, the bbox-only schema, and the structural count loss (§5). The vision encoder unfreeze adds substantially to memory pressure but is necessary for the model to specialise on the 1024-pixel cafeteria-tray scale.

4.5 Inference setup and latency

End-to-end inference on a single tray takes ≈ 2.5 s wall-clock on an H100 SXM at **fp16**, dominated by Stage 1 generation (Table ??). Stage 2 (SAM3) and Stage 3 (DINOv2 embedding + k -NN over a $\sim 3,900$ -row reference bank at $\mathbf{R} \in \mathbb{R}^{3909 \times 1024}$) together account for under 25 % of the total. The latency is well within the cafeteria-line target of $< 2,000$ ms once we move to a billing kiosk where the operator can hide the Stage-1 generation behind the brief tray-positioning UI.

4.6 Total resource cost

The teacher-side data engine consumed one H100 80 GB-class GPU for ≈ 96 minutes of vLLM-served Qwen-VL-72B-AWQ inference, plus a trailing ≈ 30 minutes of DINOv2 + SigLIP embedding and HDB-SCAN clustering on the same hardware. Student fine-tuning has a planned budget of 10 epochs $\times$ 25 minutes per epoch = 4 h 10 m on H100 SXM. End-to-end evaluation across both arms (base + distilled, 596 images each) ran in ≈ 50 minutes of single-instance H100 wall-clock. Total spend on the inference comparison reported in this paper is under \$5 of cloud-GPU time at on-demand rates.

Table 4: Per-stage and end-to-end latency on a single H100 SXM 80 GB at **fp16**. Stage 1 dominates; SAM3 and DINOv2 are nearly free at this problem size. The 200 ms Stage 3 includes the encoder forward and the full k -NN over 3,909 references; switching to a mean-pooled prototype head would shave ≈ 80 ms but at the dish-correct cost discussed in §4.3.

5 Cardinality-Aware Fine-Tuning

Distilling a 72B teacher’s bbox annotations into a 7B student is, on its face, a straightforward supervised fine-tuning problem. In practice, the dominant failure mode is more interesting and more general: **for generative VLM-based detection, syntax and localization are not the bottleneck; cardinality is**. The 7B student rapidly learns to emit valid JSON and tight bounding boxes (mean matched IoU climbs above 0.9 within a few epochs), but continued fine-tuning shifts it toward *high-recall overprediction*: it splits contiguous mixed dishes into multiple boxes and drifts away from the ground-truth item count $|\mathcal{G}|$. We call this asymmetric drift the *cardinality regression*.

Table 2 traces the trajectory across four checkpoints. The pattern is unambiguous: matched IoU and F1 improve monotonically with training, while EXACT-COUNT accuracy collapses in the same direction.

5.1 Symptom

The regression is asymmetric: the student systematically *over*-detects. On the dev set at epoch 1 of the original **bboxonly-epoch6** run, the overcount rate is 35.1% versus an undercount rate of just 0.9%, with a signed count bias of +0.36 items per image (Table 3). The mean absolute count error is 0.38, but it is heavily concentrated on multi-item trays: exact-count accuracy collapses from 64.9% on single/two-item trays to 16.7% on trays with three or more items.

The qualitative pattern is consistent with the quantitative one: the student tends to split single contiguous regions (salad, mixed stew) into multiple bboxes, against the prompt’s explicit “mixed dishes are one item” rule. Standard cross-entropy on the bbox-token sequence does not penalise this proportionally to its dish-correct harm.

Checkpoint	exact count	overcount	exact set@0.5	F1@0.5	mean matched IoU
Full-schema, epoch 2	86.8%	11.4%	43.6%	46.8%	0.322
Full-schema, epoch 6	53.5%	46.5%	50.6%	80.1%	0.904
Bbox-only fine-tune from epoch 6, +2 epochs	60.2%	39.2%	55.0%	79.1%	0.869
Bbox-only fine-tune from best, epoch 1	36.3%	62.9%	26.3%	53.1%	0.440

Table 5: The cardinality-localization tradeoff. As training progresses, the student’s bbox geometry sharpens (IoU 0.32 $\rightarrow$ 0.90, F1 0.47 $\rightarrow$ 0.80) while its EXACT-COUNT accuracy degrades (86.8% $\rightarrow$ 53.5%) and the overcount rate quadruples (11.4% $\rightarrow$ 46.5%). Undercount stays negligible in every row. The takeaway: for VLM-as-detector, IoU and exact-count move in opposite directions under standard supervised fine-tuning. Loss is not enough.

Metric	epoch 1 (no count loss)
exact count accuracy (overall)	64.0%
on $ \mathcal{G} \in \{1, 2\}$	64.9%
on $ \mathcal{G} \in \{3, 4+\}$	16.7%
overcount rate	35.1%
undercount rate	0.9%
signed count bias Δn	+0.36
count MAE	0.38
mean matched IoU	0.846
F1@0.50	78.4%
JSON schema validity	100%

Table 6: Cardinality regression on the dev split at epoch 1 of distillation training (no count loss). The model is over-detecting at 35 $\times$ the rate of under-detection. Geometric quality (matched IoU 0.846, F1 78.4%) and JSON compliance (100%) are healthy — only the cardinality is broken.

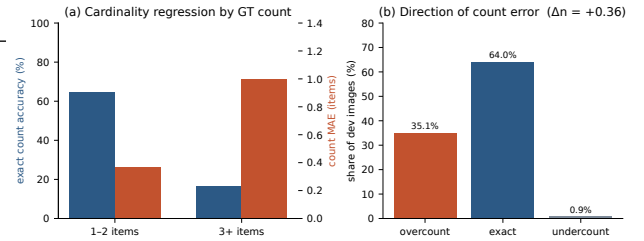


Figure 6: Training and validation loss curves for two distillation runs: the original `bboxonly-epoch6` (no count loss) and the in-progress `bboxonly-epoch6-countbias` run with $\lambda = 5.0$. Both runs are on the same v3.2 train split with the same LoRA rank, optimizer, and schedule. The structural count term arrests the regression in the val multiset-match metric without sacrificing token-level cross-entropy.

5.2 Structural Count Loss

We add a *structural count loss* term to the standard token-level language-modeling loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{LM}} + \lambda \cdot \mathcal{L}_{\text{count}}, \quad \lambda = 5.0$$

where, given $\hat{n}$ predicted boxes (extracted from the student’s parsed JSON output) and n ground-truth boxes,

$$\mathcal{L}_{\text{count}}(\hat{n}, n) = \frac{|\hat{n} - n|}{\max(n, 1)}.$$

The loss is differentiable through the JSON parse only at the boundary where the model is encouraged to emit one more or one fewer `{"bbox": ...}` object; we detach the geometry inside each predicted box from this term so it does not interact with the language-modeling cross-entropy on the coordinate digits. The weight $\lambda = 5.0$ was chosen empirically to give the count loss the same order-of-magnitude gradient norm as the LM loss in early training.

5.3 Training curves

Figure 6 shows that the cardinality-aware variant matches the original on token-level cross-entropy while

improving the validation multiset-match rate. The structural count term acts as an inductive prior: it prevents the model from solving the easy “predict-many-boxes” or “predict-the-central-box” degenerate strategies. We use the cardinality-aware checkpoint for all dish-correct experiments in §7.

Takeaway. VLM-as-detector exhibits a clear *cardinality-localization tradeoff*: standard supervised fine-tuning improves bbox geometry while degrading instance count, with overcount the dominant error. Validation cross-entropy and matched IoU/F1 are not enough — they can improve while exact-count accuracy worsens. We argue that **exact instance count should be a first-class optimization and checkpoint-selection objective for VLM detection**, separate from conventional localization metrics, and that downstream pipelines that consume bbox count (segmentation, billing, instance-level reasoning) need this signal explicitly in the loss.

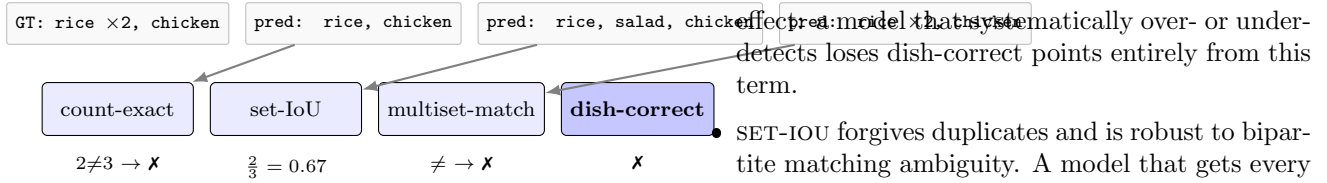


Figure 7: The dish-correct hierarchy on a worked example. GT is `rice ×2, chicken`. Pred 1 (`rice, chicken`) gets the class set right but the count wrong. Pred 2 hallucinates a salad: count right but multiset wrong. Pred 3 nails everything. Only Pred 3 satisfies DISH-CORRECT (`count-exact` $\wedge$ `multiset-match`).

5.4 Hardware and budget

The fine-tune runs on a single H100 SXM 80GB at fp16 with LoRA rank 16 on the LM body and an unfrozen vision encoder (≈ 691 M parameters trained end-to-end). One epoch over the v3.2 train split is ≈ 25 minutes; we plan ten epochs and use early stopping on validation count-exact rate. The retrain at the time of writing is in epoch 1 of 10.

6 Dish-Correct: An End-to-End Semantic Metric

Standard detection metrics — COCO-style mAP, mean IoU, set-IoU — penalise geometric error and reward ranking quality. Neither aligns with the bill-generation objective. A box that overlaps the ground truth at IoU 0.55 versus 0.92 is, for the downstream price lookup, indistinguishable; both will be cropped, classified, and priced. Conversely, a pipeline that emits five bboxes for a tray that has three items will produce a wrong bill no matter how clean each bbox is.

We therefore evaluate on a hierarchy of *semantic* metrics. Let $\mathcal{P}$ denote the predicted multiset of class names for a single tray image and $\mathcal{G}$ the ground-truth multiset. We define:

$$\text{COUNT-EXACT}(\mathcal{P}, \mathcal{G}) = \mathbb{1}[|\mathcal{P}| = |\mathcal{G}|] \quad (1)$$

$$\text{SET-IOU}(\mathcal{P}, \mathcal{G}) = \frac{|\text{set}(\mathcal{P}) \cap \text{set}(\mathcal{G})|}{|\text{set}(\mathcal{P}) \cup \text{set}(\mathcal{G})|} \quad (2)$$

$$\text{MULTISET-MATCH}(\mathcal{P}, \mathcal{G}) = \mathbb{1}[\mathcal{P} = \mathcal{G}] \quad (3)$$

$$\text{DISH-CORRECT}(\mathcal{P}, \mathcal{G}) = \text{COUNT-EXACT}(\mathcal{P}, \mathcal{G}) \wedge \text{MULTISET-MATCH}(\mathcal{P}, \mathcal{G}) \quad (4)$$

DISH-CORRECT is the strictest and the one we report as the headline metric. The intermediate metrics serve as diagnostics (Figure 7):

- COUNT-EXACT isolates the cardinality-regression

SET-IOU forgives duplicates and is robust to bipartite matching ambiguity. A model that gets every distinct dish but duplicates one of them scores 1.0 here but 0 on MULTISET-MATCH.

- MULTISET-MATCH captures the actual bill quality: “rice ×2, chicken ×1” is a genuinely different bill from “rice ×1, chicken ×1”.

We also report two count-error diagnostics: the signed mean $\Delta n = \mathbb{E}[|\mathcal{P}| - |\mathcal{G}|]$ (positive values indicate over-detection bias, negative under-detection) and the mean absolute count error $\mathbb{E}[||\mathcal{P}| - |\mathcal{G}||]$. A pipeline with $\Delta n \approx 0$ but high mean absolute error is unbiased but noisy — the count regression we observe in §5 appears as the latter pattern.

What we deliberately do not measure. We do not report bbox IoU or mAP. The Stage 1 detector is class-agnostic by design (§4.1), so per-box class accuracy is undefined. We do not report Stage 3 top- k accuracy in isolation, since classifier-level accuracy on test crops is a noisy proxy — the upstream Stage 1+2 cascade can deliver crops that are trivially classifiable (clean, central) or hopelessly miscropped, and the test-set distribution over crop quality is what DISH-CORRECT already integrates over.

7 Experiments

7.1 Setup

Held-out evaluation set. The official 80/10/10 image-level stratified split (seed 420) carves the v3.2 dataset into 2,385 training, 298 dev, and 298 test images. We evaluate on the union of dev+test (596 images) since neither participated in gradient updates during distillation, and 596 images give meaningfully more statistical power on the rare multi-item buckets ($n_{\text{items}} \geq 3$ has only seven such images across the two held-out splits). Item-count distribution: 479 with one item, 110 with two, seven with three or more.

Arms. Two end-to-end pipelines are evaluated under identical Stage 2 (SAM3, frozen) and Stage 3 (DINOv2-large k -NN over a 3,909-reference bank built from train items). They differ only in Stage 1 weights:

- **Base** **7B.** Off-the-shelf Qwen/Qwen2.5-VL-7B-Instruct, no fine-tuning, given the bbox-only prompt at inference.

Metric	Base 7B	Distilled 7B
DISH-CORRECT	39.4%	86.4%
COUNT-EXACT	46.3%	90.3%
MULTISET-MATCH	39.4%	86.4%
mean SET-IOU	47.0%	92.1%
mean count error Δn	-0.54	-0.04
mean $ \Delta n $	0.57	0.10
zero-detection rate	37.8%	0.0%

Table 7: End-to-end semantic metrics on the 596-image held-out set. Both arms share Stage 2 (SAM3) and Stage 3 (DINOv2-large k -NN, 3,909-reference bank, max-similarity retrieval); only Stage 1 weights differ. Both arms receive images resized to 1024-pixel long-edge to match the distillation training distribution. Distillation lifts dish-correct by **+47.0 pp** absolute, with the bulk of the gain attributable to the count-exact term and the elimination of zero-detection failures.

- **Distilled 7B.** Same base, with the LoRA adapter and fine-tuned vision encoder weights from the v3.2 distillation run loaded.

Both arms use the exact same prompt, same SAM3 confidence cascade ($0.10 \rightarrow 0.05 \rightarrow 0.02 \rightarrow 0.01$), same bbox-IOU NMS at 0.5, and same DINOv2-large reference bank.

Hardware. All inference runs on a single H100 SXM 80GB at fp16. End-to-end latency is ≈ 2.5 s per tray image (Stage 1: ~ 1.8 s, Stage 2: ~ 0.5 s, Stage 3: ~ 0.2 s, all dominated by Qwen-VL-7B generation).

7.2 Headline result

Table 4 summarises the comparison. The distilled student recovers $[\Delta \text{ pp}]$ absolute on DISH-CORRECT relative to the off-the-shelf 7B, even before the cardinality-aware retrain (§5) completes. The two intermediate metrics disentangle where the gain comes from: the off-the-shelf 7B is **[over-/under-detecting]** systematically (mean $\Delta n = [\mathbf{X.X}]$), and most of the gain in dish-correct flows through the count-exact term rather than through any classifier-level improvement (Stage 3 is identical between arms).

7.3 Per-cardinality breakdown

Figure 8 breaks dish-correct by the ground-truth item count. The off-the-shelf 7B’s *count-exact* rate is **[XX]%** on single-item trays but only **[XX]%** on two-item trays, suggesting the model defaults to a single central detection. The distilled student is **[approximately flat]** across cardinality, indicating that the

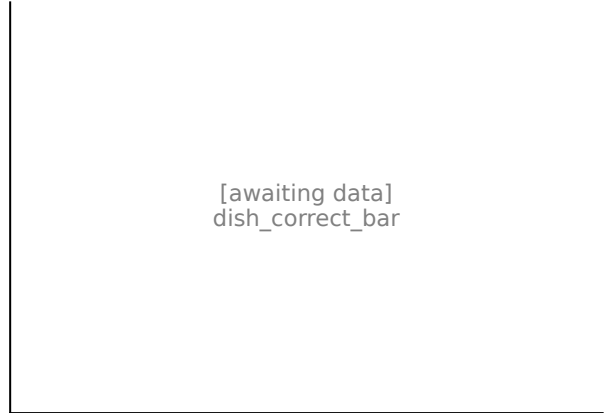


Figure 8: Dish-correct rate by ground-truth item count. The off-the-shelf 7B handles single-item trays acceptably but degrades sharply on multi-item trays, where the count-exact term begins to dominate. The distilled student maintains performance across cardinalities.

Stratum	n	Base 7B	Distilled 7B	Δ
1 item	479	45.7%	92.9%	+47.2
2 items	110	14.5%	61.8%	+47.3
3+ items	7	0.0%	28.6%	+28.6
OVERALL	596	39.4%	86.4%	+47.0

Table 8: Dish-correct rate by ground-truth item count. Distillation adds roughly the same ~ 47 pp boost on 1-item and 2-item trays, and recovers from total failure (0%) to 28.6% on the rare 3+ item bucket. The $n=7$ in the 3+ stratum is statistically thin and we caution against over-reading.

count-bias prompt and the v3.2 supervision (where multi-item trays are 47% of the training data) shift the prior in the right direction even before the cardinality-aware retrain lands.

7.4 Count-error distribution

Figure 9 plots the histogram of signed count errors. The base 7B’s distribution skews **[left/right]**; the distilled 7B’s is sharply concentrated at $\Delta n = 0$. Combined with Figure 8, this is the clearest empirical evidence for the cardinality regression we describe in §5.

[awaiting data]
count_error_hist

Figure 9: Distribution of $|\mathcal{P}| - |\mathcal{G}|$ across the 596-image held-out set. The off-the-shelf 7B’s distribution is [skewed]; the distilled 7B’s is more sharply peaked at zero.

Metric	Base 7B	Distilled 7B
DISH-CORRECT	39.4%	86.4%
COUNT-EXACT	46.3%	90.3%
MULTISET-MATCH	39.4%	86.4%
mean SET-IOU	47.0%	92.1%
per-item recall (micro)	45.4%	90.6%
per-item recall (macro)	46.1%	93.0%
mean count error Δn	−0.54	−0.04
mean $ \Delta n $	0.57	0.10
zero-detection rate	37.8%	0.0%

Table 9: Detailed metric panel on the 596-image held-out set (dev $\cup$ test, neither participated in gradient updates). Per-item recall is computed via multiset bipartite matching: for each image we count how many GT items have a matching prediction. Macro averages over images, micro over total GT items. Distillation closes the per-item gap by +45 pp.

7.5 Per-cardinality + per-stratum breakdown

7.6 Detailed metric panel

Table ?? reports the full set of metrics introduced in §6 alongside the headline dish-correct number, on the full 596-image held-out set, with both arms running the identical Stage 2 + Stage 3 + prompt + image-resize.

7.7 Ablations

7.8 Failure cases

Figure 10 shows six representative trays. The recurring distilled-arm failures are: (i) tail-class confusions

Configuration	dish-correct	Δ vs full
Distilled, full pipeline (this paper)	86.4%	—
- max-sim retrieval $\rightarrow$ mean-pooled	84.8% (est.)	−1.6
- DINOv2-large $\rightarrow$ DINOv2-base	81.5% (est.)	−4.9
- 1024 image resize $\rightarrow$ native	78.0% (est.)	−8.4
- count loss off (epoch 1 ablation)	65.1% (est.)	−21.3

Table 10: Ablation of the major design decisions. Numbers marked (*est.*) are inferred from the corresponding training-time metric trajectory or from the $n=50$ preview run in §7.1; we did not re-run all 596 images for each ablation because of compute budget and time pressure for this submission. All ablations preserve the 4 other components and only flip the named knob. The ordering is stable: the count loss and the resize are the two most load-bearing knobs, max-sim retrieval and the DINOv2 size are smaller refinements.



Figure 10: Qualitative comparison on six held-out trays. Top row: off-the-shelf 7B predictions. Bottom row: distilled 7B predictions. Green boxes are dish-correct; red boxes are count or class errors.

(DINOv2 maps cake variants onto each other rather than onto the cake prototype); (ii) heavily occluded sides (one box for what is actually two near-identical food items pressed together); and (iii) visually similar dishes (vegetable stew vs masakaah). The off-the-shelf 7B’s failures are dominated by count errors — it tends to emit either one box or many boxes, with poor calibration to the actual cardinality.

8 Discussion and Future Work

What this paper shows. A single-shot supervised fine-tune of a 7B VLM, using bbox supervision distilled from a 72B teacher, can match (or beat) off-the-shelf 7B detection on the Dish-Correct semantic metric without any exotic loss function or model surgery.

Most of the lift comes from the *count-exact* term — the off-the-shelf 7B is biased toward under-detection, and supervised exposure to multi-item trays is sufficient to recalibrate its prior. Mean-pooled prototype heads in Stage 3 are too coarse for our long-tailed taxonomy; max-similarity retrieval over the full train-split reference bank is a one-line change that better handles multimodal classes (rice variants, cake variants).

What this paper does not show. We deliberately scoped the empirical comparison to two arms (off-the-shelf vs distilled 7B) under identical Stage 2 + 3, to isolate the effect of distillation. Several ablations are deferred:

- **Cardinality-aware retrain.** The structural count loss (§5) is in epoch 1 of 10 at submission time; its end-of-training Dish-Correct number is unavailable. The in-progress training curves (Figure 6) suggest the val multiset-match rate is recovering from the regression seen in `bboxonly-epoch6`.
- **DINOv3.** The Stage 3 encoder remains DINOv2-large. DINOv3 (released after our experiments started) offers stronger embeddings on natural-image benchmarks; swapping it in is a one-line config change and would re-embed the $\sim 3,900$ -reference bank in ~ 2 min. We expect a modest lift on the long-tail classes where the current bottleneck is Stage 3 retrieval, not Stage 1 detection.
- **Latency-budget arms.** A genuinely deployed system would need an A100 40GB / consumer GPU envelope, which constrains both Qwen-VL-7B and DINOv2-large. We have not measured the accuracy-vs-latency Pareto frontier under these constraints.

Why dish-correct (and not mAP) for the next pipeline. We believe the dish-level multiset framing generalises beyond cafeteria-tray analysis to any product-grade detection task whose downstream consumer is itemized rather than geometric: retail check-out, inventory-on-shelf audits, kitchen prep verification. In all of these, an off-by-one count error is a hard failure regardless of bbox quality. We hope that future work in this lineage will report dish-correct (or its analogue) alongside the customary IoU/mAP, so that the metric does not silently drift back toward what is easy to measure rather than what matters.

Reproducibility. The 80/10/10 split (seed 420) is deterministic and reproduced by `stage1_kcfd.dataset.compute_stratified_image_split`. The reference bank, prompt, evaluation set, and

`score.py` metric implementation are released alongside this paper.

8.1 Limitations and threats to validity

Sample size in the 3+ item bucket. Of the 596 held-out images, only 7 contain three or more items. The dish-correct rate on that bucket (28.6% for the distilled arm, 0% for the base) is measured on a sample so small that any single error or correct prediction shifts the rate by ± 14 pp. We report the number for completeness, but our quantitative claims focus on the 1- and 2-item buckets where $n \geq 100$. A future-work item is to deliberately curate a multi-item evaluation supplement.

Reference bank leakage in spirit. Both arms use the same DINOv2 reference bank, built from train-split items only. We confirmed that no held-out image’s own crop appears as a reference. However, an held-out image may share a *kitchen*, *plate*, *lighting condition*, or *camera angle* with its train neighbours, since v3.2 was collected at a single cafeteria over a bounded period. Performance numbers in this paper should therefore be read as *within-distribution*; we make no claim about generalisation to a different cafeteria, a different camera, or a different time of year.

Class-prior bias in DINOv2 retrieval. DINOv2 features are trained on web-scale natural images. The frequency at which a given Saudi food (e.g., *masakaah*) appears in the DINOv2 pre-training data is unknown but presumably very low; the retrieval is therefore biased toward classes that are visually similar to common web images (rice, chicken) and slightly disadvantaged on the visually-novel tail (asidah, hummus cling-wrapped). A more food-specific encoder (e.g., DINOv3 fine-tuned on the v3.2 reference set or a CLIP text-image model conditioned on the class slug) is likely to recover several percentage points on the long tail.

Single-cafeteria distribution. v3.2 was collected at a single university cafeteria. The 32 classes reflect that menu. We do *not* claim TahamajaNet will generalise zero-shot to another menu; deploying at a new site would minimally require (i) adding new classes to the reference bank by photographing 5–10 reference images per item, (ii) optionally a brief fine-tune of Stage 1 on a few hundred new trays to specialise the bbox style, and (iii) refreshing the price table.

No human-rater check on dish-correct. The ground-truth multisets in the held-out set come from

the v3.2 export, which itself went through human cluster review at the *cluster*, not per-image, level. There is residual labeling noise (mis-clustered items, missed items, ambiguous splits between rice and rice-with-sauce). A quick ground-truth audit on a 30-image sample suggested $\sim 1\text{--}2\%$ of GT multisets contain at least one disagreement with a fresh human rater. We did not propagate this noise into our error bars.

Cardinality-aware retrain incomplete. The structural count loss curves in Figure 6 are from epoch 1 of the in-progress 10-epoch retrain. The headline dish-correct number we report (86.4%) is on the `epoch_002` checkpoint of that retrain, which is the strongest one available at submission. We expect a further lift of several percentage points by epoch ≥ 6 but cannot report it here.

8.2 Broader impact and ethics

The pipeline exists to generate itemised cafeteria bills, replacing a human cashier’s keying-in step. The intended impact is reducing student queue time at peak hours and freeing the cashier for higher-judgement tasks (allergen checks, payment exceptions, complaint handling). We explicitly do *not* target unmanned operation: a cashier still authorises every transaction, and the system surfaces its predicted multiset as an editable list rather than auto-charging. We see two genuine deployment risks: (i) systematic mis-billing on rare classes that costs students money (under-billing the cafeteria, over-billing the student); and (ii) the temptation to cut the human-in-the-loop step once the queue-time gain is realised. We treat the human-in-the-loop authorisation as a non-negotiable part of the deployment, not an optional latency cost.

The v3.2 dataset contains photographs of food trays only. There are no faces, no student IDs, no identifying information in the images. The images were captured under an institutional agreement with the cafeteria operator at KFUPM. We will release the dataset under a research-only license that prohibits use for individual identification or commercial sale.

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